Do small clauses form a uniform class?: Evidence from structural priming

Some linguistic analyses suggest that seemingly unrelated constructions may nevertheless be structurally related. For example, certain embedded clauses selected by verbs like *consider* (e.g., *He considers her smart*), causative verbs (e.g., *He made her leave*), and possibly ditransitive verbs (e.g., *He gave her a book*) are sometimes analyzed as forming a syntactic natural class known as *small clauses* (see den Dikken, 2006), clauses that lack overtly tensed predicates. However, it remains unclear whether the notion of small clauses reflects a genuine mental category or merely a descriptive label. The current study tests whether speakers possess a mental category that encompasses different subtypes of small clauses, using structural priming (Bock, 1986), the tendency for speakers to reuse recently encountered structures.

A previous study by Luo et al. (2021) reported that *consider*-type small clauses primed speakers to use the Double Object construction (DO; *She gave John a book*) relative to a mono-clausal intransitive control, suggesting that *consider*-type small clauses and DO constructions belong to the same syntactic natural class. This finding is consistent with independent analyses proposing that the direct and indirect objects form a small clause (Johnson, 2018).

We conducted three recall-based structural priming experiments to replicate and extend Luo et al. (2021). In critical trials, participants read aloud and memorized two sentences and then recalled the first sentence (the target) given lexical cues. Half of the target sentences were DO and the other half were Prepositional Dative (PD), constituting the Target Type manipulation. The second sentence served as a prime and varied across conditions (Prime Type; see Table 1). In filler trials, participants occasionally recalled the second sentence instead. We coded whether participants produced DO (1) or PD (0) in their recalled responses and analyzed the data using Bayesian mixed-effects logistic regression with Target Type (DO vs. PD), Prime Type (small clause vs. non-finite control vs. finite control), and their interaction as fixed effects, along with maximally structured random effects (Barr et al., 2016).

In **Exp.1** (n = 50), small-clause primes increased DO responses by 5.9% for PD targets and 2.5% for DO targets (estimated effect of small clause vs. other primes: b = 0.60, SE = 0.30, 95% CrI = [−0.01, 1.25]). In **Exp. 2** (n = 120), which used longer target sentences to reduce verbatim recall but was otherwise identical to Exp. 1, small-clause primes increased DO responses by 8% for PD targets and 2.6% for DO targets (b = 0.36, SE = 0.15, 95% CrI = [0.08, 0.65]). There was no evidence for a difference between non-finite and finite control primes (b = −0.07, SE = 0.18, 95% CrI = [−0.47, 0.32]). Together, Exps. 1 and 2 conceptually replicate Luo et al.'s finding that consider-type small clauses prime the DO structure.

Exp. 3 (n = 80) tested whether small clauses in causative constructions also prime DOs, while simultaneously replicating the priming from consider-type small clause to DOs. As shown in Table 2, Experiment 3 included an additional factor, Clause Type (selected by consider-type verbs vs. causative verbs), fully crossed with Prime Type (small clause vs. non-finite control). In the *consider*-type conditions, speakers were on average 3.0% more likely to produce DOs following small-clause primes than following non-finite primes. In the causative conditions, the corresponding increase was only 0.4%. However, the statistical analyses did not provide clear evidence for any of these effects (Prime Type: b = 0.15, SE = 0.14, 95% CrI = [−0.12, 0.42]; Clause Type: b = −0.19, SE = 0.14, 95% CrI = [−0.47, 0.09]; interaction: b = 0.27, SE = 0.27, 95% CrI = [−0.25, 0.80]). Finally, we conducted a Bayesian analysis on the aggregated data from Exps. 1–3 to assess the overall evidence for each priming effect. Pairwise comparisons estimated the difference in DO response rates between small-clause primes and baseline primes (finite and non-finite controls) as 0.38 (95% CrI = [0.17, 0.57]) for *consider*-type constructions. In contrast, for causatives, the estimated difference was 0.02 (95% CrI = [−0.34, 0.39]).

These results show that small clauses selected by *consider*-type verbs prime DO constructions, suggesting they are structurally similar. By contrast, we found no evidence that small clauses selected by causative verbs prime DOs, suggesting that constructions describable as small clauses may not form a uniform category in speakers' mental representations.

Table 1: Prime conditions in Experiment 1 & 2

Prime Type	Example sentence
finite	The teacher considered that the student was smart.
non-finite	The teacher considered the student to be smart.
SC	The teacher considered the student smart.

Table 2: Prime conditions in Experiment 3

Clause Type	Prime Type	
causative	non-finite	The teacher aided the student to pass her class.
	SC	The teacher helped the student pass her class.
consider-type	non-finite	The teacher considered the student to be smart.
	SC	The teacher considered the student smart.

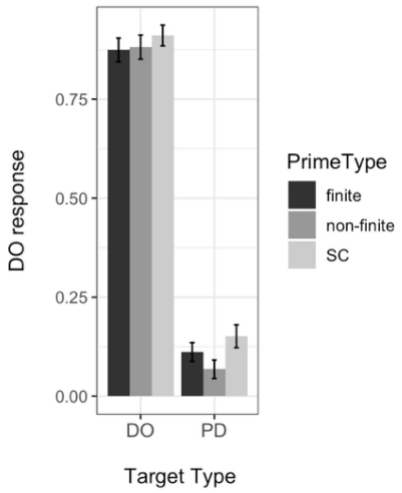

Figure 1: Results of Exp. 1

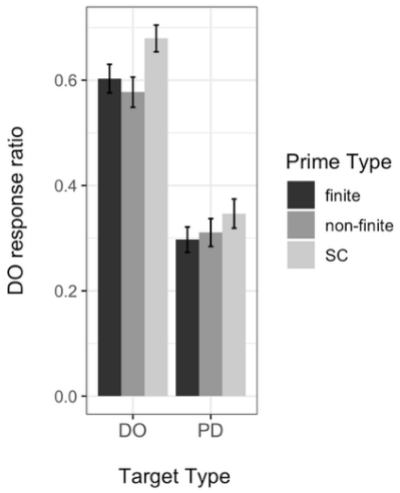

Figure 2: Results of Exp. 2

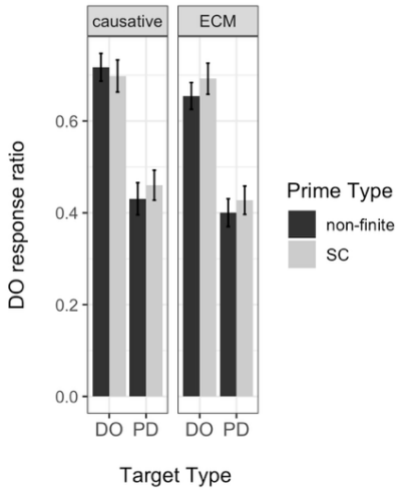

Figure 3: Results of Exp. 3

References

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